

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457925.7805 +/- 0.0049 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.117 +/- 0.047
Transit depth (Rp/Rs)^2	1.37 +/- 1.11 [%]
Orbital Inclination (inc)	89.7 +/- 2.92
Airmass coefficient 1 (a1)	1.07032 +/- 2e-05
Airmass coefficient 2 (a2)	-0.028 +/- 0.017
Scatter in the residuals of the lightcurve fit is	0.00038 %
Best Comparison Star	#2 - [196, 431]
Optimal Aperture	79.57
Optimal Annulus	203.09
Transit Duration (day)	0.087 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.117 $\pm$ 0.047 vs 0.1326 (deviation 0.33 σ)
Duration fit vs published	0.087 d vs 0.0737 d (deviation 1.1 σ)
Mid-time O-C	4.8 min
Depth SNR	2.5
Residual scatter	0.00038 %

Light curve

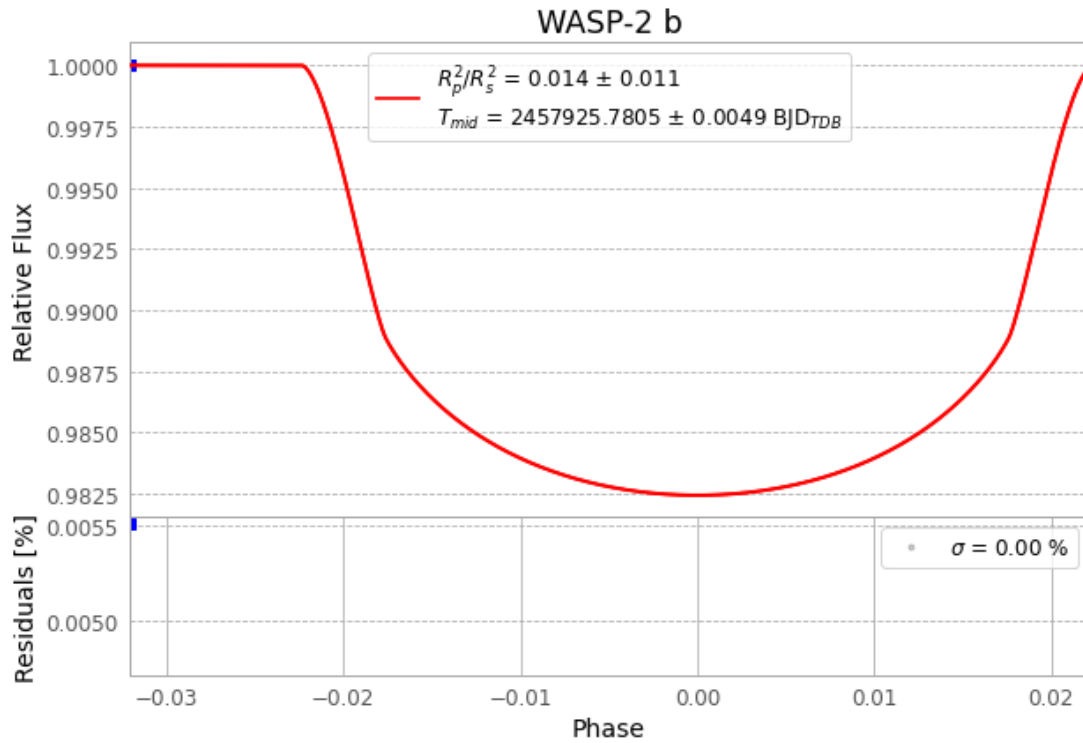


Figure 1 — FinalLightCurve_WASP-2 b_2017-06-20.png (SHA-256 3e465718c3a1393e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [81, 322], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 85f3d5fc97aa109b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62be8a4

Data provenance

83 FITS frames (55 MB), WASP-2170621045712.FITS → WASP-2170621090312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-170621.log (902162ed0696...), FinalLightCurve_WASP-2 b_2017-06-20.png (3e465718c3a1...), FinalLightCurve_WASP-2 b_2017-06-20.csv (ceacc7fda592...)

Compute resources

Wall time 1006.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 17:25Z.